

Unit 4

Common Business Application of Information Systems

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4.0 AIMS AND OBJECTIVES

After studying this unit the student should be able to:

- explain the need and uses of financial information systems
- understand the importance of marketing information systems

- describe the nature and uses of manufacturing and production information systems and;
- understand the impact of human resource information systems.

4.1 INTRODUCTION

So far, you have learned about computer information systems, the decision making process, and the basic computer resources available to the manager. What you learned generally emphasized the organizations. You will apply what you have learned about information systems, decision-making, and computer system resources to problems in finance, marketing, manufacturing and production and human resources.

4.2 FINANCIAL INFORMATION SYSTEMS

The Financial Management Function

The financial management functions of any organization encompass a number of important responsibilities. These include monitoring and analyzing the organization's financial condition through the management of accounting systems and the preparation of financial statements and reports. The financial management function is also responsible for the budgeting process, the management of customer credit, the calculation and payment of income, payroll, property, excise and other taxes; and the management of appropriate insurance coverage for the organization's personnel and assets. Furthermore, the function must provide for an auditing process to ensure the accuracy of the financial information kept and to protect the investor's funds.

The financial management function is also responsible for the management of the organization's fixed assets, pension funds, and investment in existing plants and equipment, subsidiaries, or other investments. The function also includes the evaluation of new investments and the acquisition of borrowed funds or capital to pay for them.

The financial management function must manage the cash flow of the organization in such a ways as to ensure that the organization has the necessary funds to operate, that extra cash is

not left idle, and that the organization has the borrowing power to meet its cash needs when the cash flow is insufficient.

Clearly, the financial management function is very important to the organization. Because of its importance, financial management personnel require information systems that provide them with accurate, timely and appropriate information to carry out their functions.

Like all information systems, financial information systems may be categorized into those that support operational tasks, those that support tactical decisions, and those that support strategic planning decisions.

4.2.1 Operational Financial Information Systems

Whether you are the chief operating officer of a large international organization or merely the supervisor of a small department at one location in that organization, you will find it necessary to make financial decisions. The financial decisions you may encounter as a manager are diverse and could include decisions similar to these:

Should you purchase a large new piece of equipment or lease it for a three-year period?

How much of your department's funds should you allocate to telephone usage rather than travel?

Should you invest money in new computer equipment or additional merchandise for resale?

Should you allow a particular customer to make a large purchase on credit?

Should your organization decide to offer a new product or service, or should it provide the same products and services in additional markets?

To make these and similar decisions, you need to understand the basic financial accounting systems found in most organizations. You must also understand how financial information systems can help you make improved decisions about the financial problems within your purview.

Operational Financial Information Systems

Typically, the applications first computerized by many organizations are accounting systems designed to produce operational-level output. The development of computerized financial information systems for use in tactical decision making and strategic planning is usually deferred until the basic operational-level information systems are in place and working.

Operational financial information systems produce the routine, repetitive information products that every organization finds necessary. These products, or output, include paychecks, checks to vendors, customer invoices, purchase orders, stocks reports, and other regular forms and reports. Financial operating information systems are typically task oriented. They focus on processing financial transactions to produce the financial products just described.

Accounting Systems

The heart of an organization's financial operating information system is the accounting system. A computerized accounting system is composed of a series of software modules or subsystems that may be used separately or in an integrated fashion. The subsystem modules include;

General ledger: The general ledger subsystem ties to other financial accounting systems subsystems together, provides managers with periodic account reports and statements, such as an income statement and a balance sheet, and provides support for budgeting.

Fixed assets: The fixed assets subsystem maintains records of equipment, property, and other long term assets an organization owns. The records include the original cost of the assets, the depreciation rate on each asset or groups of assets. the accumulated depreciation to date, and look value of the asset, which is the original cost less accumulated depreciation to date. The general ledger subsystem uses this information to maintain up-to-date balances in the various long-term asset accounts of the organization. The subsystem also may maintain and process data on the gain or loss on the sale of fixed assets and prepare special income tax forms for fixed assets required by the federal government.

The Sales Order Processing Subsystem: The sale order process subsystem, or order entry subsystem, routinely records sales orders and provides the documents that other subsystems use to fill those orders that maintain inventory levels and that bill the customers (sales invoices). The subsystem provides sales tax data to the general ledger subsystem for updating inventory balances, and sales invoice data to the accounts receivable subsystem for posting to customer accounts.

A computerized sales order subsystem usually tracks the sales made by each sales person and provides input to the payroll subsystem so that salesperson's commissions can be accumulated. The subsystem should also provide information to the shipping department to ensure that the correct stock is sent to the customer; provide for backorders when there is not enough stock on the shelves; accurately figure prices, totals, discounts, and taxes on the order; and allow a quick and accurate response to customer inquiries about the status of the order.

Accounts receivable Subsystem: The accounts receivable subsystem allows you to enter, update, and delete customer information, such as charge sales, credit terms, cash payments received, credit for returned or damaged merchandise, and accounts balances.

Inputs to the accounts receivable subsystem include sales invoices, credit memoranda, and cash received from customers. Typical outputs are monthly customer statements of account and a schedule of accounts receivable listing each account and its balance.

Many accounts receivable subsystems produce aged accounts receivable reports. These reports classify accounts balances into several categories (30, 60 or 90 days over due). An aging report identifies customers with overdue balances to managers, allowing them to use the computer system to prepare collection letters, start collection procedures, and disallow additional credit to poor credit risks. Without aging data, an organization may continue to grant credit to customers already long overdue on their payments.

The Accounts Payable Subsystem: The accounts payable subsystem processes much of the same routine, repetitive information as the accounts receivable subsystem, except that the information is about organization's creditors rather than customers. For example, the subsystem maintains creditor account information, prepares checks to creditors, and produces the accounts payable schedule. The accounts payable subsystem provides data directly to the general ledger subsystem and receives data from the purchase order subsystem.

The accounts payable subsystem may also find the due dates for purchases on account and the last date on which cash discounts may be taken on those purchases. This subsystem provides important operational level information that can be used to schedule cash payments to creditors. The effect

of such a subsystem is to allow the organization to keep its money working for it as long as possible and yet ensure that all cash discounts are taken.

The Inventory Control Subsystem: The inventory control subsystem provides input to the general ledger subsystem and receives input from the purchase order and the sales order subsystems. The basic purpose of the subsystem is to keep track of inventory levels and inventory costs for the organization. The subsystem maintains information about each stock item, such as stock numbers and stock descriptions, receipts and issues of stock, and stock balances. The inventory subsystem may also trigger the purchase of stock when stock levels reach certain points, maintain information about stock item costs, and maintain selling price data on each stock item.

The inventory subsystem maintains stock balance data by obtaining stock receipts from the purchase order system and stock issues from the order entry system. Updates for other stock changes, such as damaged goods, lost stock, shrinkage, or spoilage are usually entered directly into the subsystem by inventory clerks after a physical inventory has been completed.

The Purchase order Processing Subsystem: The purchase order processing subsystem processes purchase orders and tracks which purchase orders have been filled, which stock items ordered are on backorder, which stock items have been damaged or do not meet the specifications of the original order, and when orders are expected to be received.

The purchase order subsystem provides information to the accounts payable and inventory subsystems. The subsystem produces a variety of reports, including a backorder report listing all stock items on backorder and an open order report listing all purchase orders not yet received and their expected arrival dates.

The Payroll Subsystem: The payroll subsystem processes wage and salary information, such as payment to employees; deductions from employee checks; and payments to government taxing agencies for taxes owed. The payroll subsystem produces weekly payroll summary reports, overtime reports; forms for taxing agencies, such as wage and tax statements; payroll checks for payroll taxes owed to taxing agencies.

When these computerized accounting subsystems are integrated, each subsystem receives data as input from other subsystems and provides information as output to other subsystems. The accounting subsystems might be integrated.

4.2.2 Tactical Financial Information Systems

The computerization of accounting systems usually changes the way managers' view accounting information. Because a large database of information becomes available in computerized form and can easily be extracted or manipulated, this information begins to be viewed as a resource for tactical planning. Suddenly, it becomes possible for managers to get important summaries and comparisons of accounting data easily and swiftly. In the past this information would have taken a great deal of time to extract from a manual accounting system. The result is that managers view the accounting system as more than merely a producer of checks, invoices, and statements. It becomes repository of important data that can assist management in decision-making. The computerization of accounting system has helped to spawn the use by managers of corporate databases to support tactical decisions and

strategic planning. This has led to the development of computer supported financial information systems for tactical decision making and strategic planning information systems.

The Nature of Financial Information Systems for Tactical Decision Making

Tactical financial information systems support management decision making by providing managers with regular summary reports, regular exception reports, ad hoc reports, and other information that will help them

- 1) control their areas of responsibility and
- 2) Deploy their resources to pursue organization goals. Whereas operational control systems are focused on tasks, tactical information systems are focused on resource allocation.

It is possible to design many computer supported tactical level information systems for the financial decisions that managers must make. Common systems include budgeting systems, cash management systems, capital budgeting systems, investment management systems. Each of these systems will be briefly described.

Budgeting Systems

The general ledger subsystems of many computerized accounting systems permit the user to enter budget amounts by account number. Periodically, (weekly, monthly, quarterly, or annually) the budgeted amounts (allocations) and the actual amounts spent or received (actual) for each account are compared and various reports are prepared. For example, the general ledger subsystem of a mainframe accounting system may provide the reports listed below.

1. current budget allocations by line item.
2. current budget allocations compared to year to date revenues and expenditures.
3. budget variances by line item type, or the differences between allocations and actual
4. current budget allocations compared to the previous year's allocations
5. current revenues and expenditures compared to the previous year's revenues and expenditures

6. Current revenues and expenditures compared to the average of the other units or divisions of the organization.

Reports such as these may be prepared for a department, a division, a subsidiary, or the entire organization. The budgeting system permits managers to compare revenue and expense data against the standard of the budget allocations. It also allows prior fiscal period, other division or department, industry wide data to be used as standards against which current budget amounts may be compared.

Cash Management Systems

Important functions of financial management include ensuring that the organization has sufficient cash to meet its needs, putting excess funds from any period to use through investments, and providing borrowing power to meet the organization's cash needs in those periods when there is an insufficient cash flow.

There are two major reasons why an organization needs cash: for working capital (cash needed for day to day operations) and for the acquisition of long term assets. To determine if adequate cash is available for its working capital needs and its long term asset acquisition plans, the organization must prepare a report of its expected cash flow for the time periods being considered. Typically this report shows the cash flow for each month of the coming year.

A cash flow report shows the estimated amount of cash that will be received and spent each month. The report shows in which months there will be cash received and spent each month. The report shows in which months there will be excess funds that might be put to use, and in which months there will be insufficient funds, required the organization to borrow cash to meet its working capital or fixed asset acquisition needs.

The information provided by a cash flow helps the manager make decisions about investing, purchasing, and borrowing money. If this information is placed on an electronic spreadsheet, the manager may stimulate a number of possible business conditions, such as (1) increasing or decreasing revenue, (2) increasing or decreasing customer credit problems, (3) deferring the acquisition of an asset, or (4) repairing existing fixed assets instead of replacing them.

Capital Budgeting Systems

A capital budget contains information about the planned acquisition or disposal of major plant assets during the current year. The manager may wish to compare the various capital spending plans in terms of three commonly used evaluation tools: net present value, internal rate of return, and payback period. These tools can easily be calculated using computer systems. What if? games are performed using software packages.

Investment Management Systems

Investment management – overseeing the organization's investments in stocks, bonds, and other securities – is an important part of cash management. Managing investment is also an important part of an organization's pension plan management. Whatever the source of the funds for investments, most organizations invest money in securities of one kind or another. Careful management of these investments is necessary to ensure the achievements of organization goals.

Current information systems provide unique ways to manage stock and bond portfolios. These ways typically involve the use of external databases that furnish immediate updating of stock and bond prices, information about the history of each investment, and various portfolio investment analysis tools to help the manager stay on top of the organization's investments. The system may be a simple one in which the manager's microcomputer is equipped with a modem and he or she is provided with a subscription to an investment service. Possible information that can be generated are current dividend, price history, price stability index, projected changes earnings, current earnings per share, debt as a percent of capital, dividends history, industry ranking, low price per year and the like. Tactical financial information systems give the manager increased control over the financial resources of a department or an entire organization, and provide considerable support when he or she is deciding the allocation of financial resources to meet organizational goals.

4.3.2 Strategic Financial Information Systems

You have learned that operational control level information systems are task oriented and that tactical level information systems are resource allocation oriented. In contrast, strategic information systems are goal oriented. That is, these systems are concerned about goal and direction setting for organizations.

Strategic information systems typically include several types of information flows:

1. Financial condition analysis data or an analysis of internally generated information describing the status of the organization
2. Economic, demographic, and social data or an analysis of externally generated data describing the present and future environment for the organization
3. Forecasts of the future of that organization in those environments

Two major outcomes of financial strategic planning are the setting of financial goals and directions for the organization. The former may include setting investment goals and return on investment goals. The latter may involve deciding on new investment opportunities or on the mix of capital sources used to fund the organization.

A major source of computerized information about the current and future status of the organization is the organization's own financial accounting database. A promising source of computerized information pertaining to the present and future environment in which the organization must operate are on line databases for economic, social, demographic, technological, and political information. Projecting likely scenarios for the organization using these two categories of data is the art of forecasting. A major purpose of strategic decision-making is to use long-range forecasts to reduce the risk involved in major organization decisions.

Financial Conditions Analysis Systems

Computerized accounting systems provide the user with a variety of reports on which many ratios and analysis tools may be applied.

Example, working capital = current assets ñ current liabilities.

Long Range Forecasting Systems

Strategic planners demand forecasts on a variety of factors that will affect organization performance in the future.

- Forecasting the sales revenue of a potential new product makes important information available to planners considering the development and marketing of the new the product.

- Forecasting the financial health of the organization through the long range budget estimates ñ including a variety of possible wage negotiation settlement, actions by competitors, interest rate fluctuations, fuel costs changes, and different inflation rates provide planners with opportunities to consider actions that will help the organization survive bad times or take advantage of a future environment.
- The types of information used in forecasting the future environment of the organization are many and diverse. They include descriptions of the present economy and forecasts of the future economy, information on the present demographic structure of the region or country and forecasts of the demographic structure of the region or country, and descriptions of the current social structure and social mores and predictions on the future structure of society and societal mores.

4.3 MARKETING INFORMATION SYSTEMS

Marketing information systems support the major activities of a marketing organization. The information systems collect data that describe marketing operations, process those data, and make marketing information available to marketing managers to help them make effective decisions.

4.3.1 Operational Marketing Information Systems

Operational marketing information systems primarily produced routine, repetitive, expected, and regular data that describe past marketing activities. The information they produced is usually detailed, highly structured, and accurate, and is derived from internal sources.

Sales Information Systems

Salespeople are responsible for many sales activities. They must identify potential customers, make the customer contact, call on the customer, make the sales, close the sale, and follow up on the sale. These are the bread-butter sales activities of the salesperson. There are many information systems that support the salesperson in these activities. Among these information systems are the following:

Prospect Information System

Locating potential customers is often a time consuming and frustrating part of the salesperson's work. The sources of information used to obtain leads about prospective customers are frequently diverse and may include other customers, other vendors who sell supporting or ancillary products, newspapers notices, telephone directories, and direct customer inquiries. The search of directories and other lists of customers may be very time consuming and yield few customers.

Files of sales leads are often called prospect files. When these files are stored on magnetic media, they are easier for the salesperson to search or summarize. Outputs of prospect information systems may include lists of prospects by location, by product category, by gross revenue, or by other classifications important to the sales force.

Contact Information Systems

Customer contact information systems provide information to the sales force pertaining to customers, their product or service preferences, sales history data, and a historical record of sales calls and/or visits. One output of these systems may be a call report showing the number of sales calls made by a salesperson categorized by size of organization, previous sales, or some other characteristics, and the number of sales made per customer, per visit, and or per category.

Inquiry Information Systems

When customers make inquiries about the products and services the organization offers, the inquiries need to be processed, recorded, and stored for analysis or for sales contact. It is important that inquiries be associated with the actual or potential customer who made the inquiry, what products or services the query pertained to, when the inquiry was made, and where the potential customer was located. It is also very important to record these data on a medium that will allow analysis easily at some future time.

Document Information Systems

- Sales form letters for salespeople so that they can mail well written, clear and effective letters

- Quotation documents are frequently produced. A document information system provides marketing personnel with *ready-to-adapt documents* for their operations. The system also improves the quality of the documents developed by marketing personnel, and is likely to increase sales revenue.

Order Entry Systems

A great deal of marketing data is created through the financial accounting system of the organization. For example, the order entry system provides the marketing manager with the raw data from which to obtain gross sales by time period, salesperson, product, and territory. This information may be used for a variety of marketing decisions at several marketing decision levels. Analysis of product sales through analysis of the sales orders provides the marketing managers with some of the information on which sales forecasts can be based, for instance.

Telemarketing Systems

Using the telephone for selling is referred to as telemarketing. Using telemarketing has become a very important means by which companies have improved the productivity of their sales force. Using the telephone to initiate contacts, offer products and services, follow up on sales eliminates travel costs and travel time and lets salespeople reach many more customers in a given time period than they could have through conventional means.

Some telemarketing systems include computer support for the automatic calling of parties and/or the automatic delivering of a voice message to the answering party under the control of computer system.

Direct Mail Advertising Systems

Many firms generate sales by mailing sales brochures and catalogs directly to customers. To distribute sales documents rapidly to large numbers of potential customers, most marketing departments maintain customer mailing lists that are used for mass mailings. The lists may be drawn from customer files, accounts receivable records, or prospect files, or they can be purchased from other documents.

Inventory Information Systems

Inventory subsystems of the financial accounting system provide information about inventory levels, stock outs conditions, stock receipts, stock issues, stock damage, and the location and distribution of stock within the organization. For example, salespeople should be able to check the amount of available stock in inventory before closing a sale with a customer. Thus, the inventory information system provides important data regarding stock levels to the order entry system.

Distribution Information Systems

If the organization uses its own distribution system, information about its effectiveness must be collected and reported to management. Information should also be maintained about the acquisition, repair, use and allocation of equipment.

Credit Information Systems

An operational level credit information system usually provides the salesperson or credit manager with information pertaining to the maximum allowable credit that should be granted to a customer. This information is ordinarily integrated with the order entry subsystem of the financial accounting system. The credit information system may manage information on the credit history of customers; this data can include information from external sources, such as credit agencies.

4.3.2 Tactical Marketing Information Systems

Tactical information systems differ from operational marketing information systems in that they may provide produce information not only on a regular basis but also generate ad hoc reports, create unexpected as well as expected output, produce comparative as well as descriptive information, provide summary information as opposed to detailed data, included both internal and external data sources, and processes subjective as well as objective data.

Examples of tactical marketing information systems that will be discussed in this part of the paper include those that support the development and management of the sales force, the management of advertising and promotion campaigns, the distribution and delivery of products sold, and product pricing.

Sales Management Information System

A major objective of sales manager is to reach the sales goals set by top management. To accomplish this objective, sales managers must make many tactical decisions, such as sales territories should be shaped, how the sales force should be allocated within those territories, and what emphasis should be placed on the products offered and customers served. Sales managers must decide how to reward salespeople to encourage increased sales efforts, which market segments should be emphasized to best reach sales goals, and which products and services will best appeal to each segment. They also must monitor the progress of the sales effort to determine if their decisions were correct or if they need to change their tactical plans.

To make these decisions effectively, sales managers should have at their disposal a great deal of data about the sales histories of each salesperson, territory, product, and market segment. Managers can use these data ñ provided by sales management information systems ñ to develop reports analyzing sales activities that help them make decisions about salespeople, territories, products, and customers. Analysis of past sales efforts might reveal, for instance that the greatest volume of sales is obtained when certain products are emphasized with certain market segments. This information may be obtained from a report that correlates product or service categories with customer categories.

Through the use of these reports, marketing managers may adjust advertising media, promotion schemes, sales calls, and sales approaches to emphasize selected products to very narrowly defined market segments to maximize gross sales.

In addition to planning future campaigns, sales managers need information to control current campaigns. The manager monitoring sales progress may wish to recursive daily, weekly, or monthly reports on the sales achievement of each salesperson with averages calculated for the entire sales force. This information might be used to evaluate salesperson effectiveness, the success of the sales reward system, or the appropriateness of the shape of the territories assigned. A report showing the percentage of each salesperson's sales by market segment and by product along with averages for the total sales force would display the current effectiveness of each salesperson in selling the organization's products to each customer group.

The manager can compare sales, product, customer, and territory information from one department against external benchmarks of success, such as the organization as a whole or industry averages. The manager, for example, may wish to compare staff deployment to the deployment of marketing staffs in other marketing departments within the organization or to the industry as a whole. The marketing manager might also view the amount of money spend in the department on salespeople versus support staff and then compare those amounts to the amounts spend by other marketing departments in the organization or to the industry as a whole. If sales data are added to expense data, the marketing manager can compare the productivity of the industry. Comparisons that indicate poor productivity might lead the marketing manager to compare the productivity of each salesperson to the average for the department. This comparison, in turn, might lead to decisions changing the configuration of sales territories, training for specific salespeople, product emphasis for certain territories, market segment emphasis for certain sales personnel, or a variety of other remedies.

These examples should not imply that the marketing manager's job can be performed merely by projecting historical information on current problems. If all marketing decisions could be made merely by referring to the past performance of salespeople with customers and products within territories, marketing managers would not be needed at all. The marketing manager must bring to the table not only historical information, trends, and probabilities, but also considerable knowledge of the current environment. In the end, the marketing manager must make decisions based on less than all of the information desirable. Marketing information systems do not make the decisions for marketing managers, they provide information to support those managers in the decision making process.

Advertising and promotion information systems

Advertising and promotional tactics also need to be developed by marketing managers to implement strategic sales goals set by top management. Managers must decide which advertising media and promotional devices to use to reach the selected market segments, when these media and devices should be used, and what overall mix of promotional activities should be deployed to achieve sales goals. Advertising and promotion information systems assist managers in these tasks.

To decide what advertising and promotional tactics to use, marketing managers need such information as market segment history, the effectiveness of previous advertising and promotional efforts on each market segment, and the sales history of products by market segment. Historical data on the effectiveness of various advertising and promotional instruments and the sales of products by market segment are not foolproof methods for forecasting future success. Past profiles and trends may not necessarily be projected into the future. The areas of advertising and promotion contain too much managerial art for projection of past data to be perfectly successful. However, the past success of specific advertising and promotion campaigns for specific products and customers is very important information for the manager.

Production Pricing Information Systems

Product pricing information systems are important because the price of a product or service affects the sales volume and profitability of the organization. While the marketing manager usually selects a price that will at least recover production costs, the price chosen is constrained by the prices of competitors for similar products or services and for alternative products or services. To make pricing decisions, the marketing manager should have information pertaining to the expected demand for the product or similar products, the desired profit margin for the organization, the costs of producing the product or providing the service, and the prices of competing products.

To help marketing manager price products and services, some organizations develop a pricing model that allows him or her to input data on a variety of forces affecting prices such as expected competitive prices, expected consumer price indexes, expected consumer disposable income, volume of expected advertising expenditures. The model then uses arithmetic algorithms that represent the organization's assumptions about the interrelationship of each of these factors. This makes it possible for the manager to play with the inputs to identify the recommended price for a product under a variety of possible conditions.

Distribution Channel Decision Support Systems

Deciding how the organization's products and services will be delivered to the ultimate consumer is another important tactical decision that must be made. The organization needs to determine if it wishes to sell all or some of its products and services directly to its customers,

use middlemen, or both. If direct sales are considered, then the choice of trade channels or channels within that delivery mode – for example, sales force, direct mail, telemarketing, and so on – must be decided on. If middlemen are chosen to distribute the product or service, the choice of the channel will be affected by how well the channel distributes the products or services among the organization's targeted market segments.

To support the marketing manager, the marketing information system should provide a distribution channel decision support systems. This system should provide information such as the costs of using the various distribution channels, the time lags caused by the various channels, the reliability of the various channels in delivering the products and service, and the marketing segment saturation provided by the channels.

4.3.3 Strategic Marketing Information Systems

To develop an overall marketing plan, an organization may engage in a variety of tactical and strategic planning activities. The strategic activities include segmenting the market into target groups of potential customers based on common characteristics or needs or wants, selecting those market segments the organization wishes to reach, planning products and services to meet those customers' needs, and forecasting sales for the market segments and products.

Sales Forecasting Information System

Strategic sales forecasting usually includes several varieties of forecasts: forecasts of sales for the entire organization, forecasts of sales for each product or service, and forecasts of sales for a new product or service. The results of these sales forecasts will often be further categorized by sales territory and sales division. Regardless of the type of forecasts, sales forecasts are usually based on more than historical data; they are not merely projections of past trends. Sales forecasts are also based on assumptions about the activities of the competition, government action, shifting customer demand, demographic changes and movements, and a variety of other pertinent factors, including even the weather. Errors in sales forecasting will have many implications in the other aspects of the organization.

Product Planning and Development Information Systems

The major objective of *product planning and development information systems* is to make information about consumer preferences obtained from the marketing research system

available for the development of new products. The primary output of planning and development activities is a set of product specification. The specifications will serve as inputs for other decisions such as product design.

4.3.4 Tactical and Strategic Marketing Information Systems

Two important information gathering systems provide support for both tactical and strategic marketing decisions. These two information systems are marketing research systems and information systems that collect data about the organization's competitors.

Marketing Research

In large organizations, research departments conduct and manage marketing research. In small companies, marketing research may be completed by outside consultants or by personnel who must wear several hats. Regardless of how the function is completed, the results of marketing research provide important inputs to tactical and strategic decision making.

Inputs to marketing research are heavily derived from sources external to the organization. These inputs include such widely diverse sources of customers as customers, potential customers, census and demographic data, industry or trade data, economic data, social trend data, environmental data, and scientific and technological data. These data may be obtained through such means as direct mail surveys of customers, personal and telephone interviews of consumers, library searches of governmental and industry reports, searches of the databases of information utilities, and reports filed by sales personnel.

Marketing research personnel make heavy use of statistical methodology in analyzing the data collected and in reporting the information to the organization. Obtaining totals, counts, and averages in terms of consumer responses to questions, correlating social and economic characteristics of customers with their buying practices, completing times series analysis of past industry wide sales to determine the projected sales of a product, and testing hypothesis about consumer response to differing product packaging represent only some of the statistical procedures that are used to analyze information for marketing managers.

Typical of the activities of a marketing research department are:

1. conducting trend analysis of the sales of products and services identical or similar to those offered by the organization to identify products or services that are on the ascent or descent
2. analyzing population and target group characteristics, especially for trends or changes in data that could affect the organization
3. analyzing and identifying consumer preferences, including the testing of products and services
4. determining and analyzing customer satisfaction with the organization's existing products and services
5. estimating market share for all or each product and service offered.

The results of marketing research are often presented graphically, in the form of tables, charts, and graphs.

Competitive Tracking Information Systems

To ensure that the marketing mix offered by your organization will continue to be effective in satisfying customers, you must keep abreast of major competitors and their marketing activities. Market share is likely, in the end, to be the greatest for the organization that provides the marketing mix most closely matching a given market segment's needs and wants. Thus, knowledge of competitor prices, products, sales, advertising, and promotions must be gathered if the organization is to avoid falling behind the competition in the eyes of the customers. This task is carried out through *competitive tracking information systems*.

Information about competitor activities is also obtained from informal sources, through such activities as reading trade journals and newspapers, visiting competitor distribution outlets, and talking to competitor offers and employees at conventions. Salespeople in most organizations are encouraged to provide feedback about competitor activities by filing field reports. The information they obtain may be gathered through observation of competing salespeople or by seemingly casual questions that addressed to customers. Information about the competition may also be gathered more systematically by conducting keyword searches in external databases or information utilities.

4.4 MANUFACTURING AND PRODUCTION INFORMATION SYSTEMS

Purpose: manufacturing systems encompasses all the activities necessary to ensure production. These activities may include the evaluation of sites for production; the planning, development, and maintenance of production facilities; and the setting of production goals to meet the requirements of the sales forecast generated by the marketing system.

Manufacturing and Production Information Systems

Manufacturing and production information systems provide the data necessary to plan, organize, operate, monitor, control and otherwise manage production systems.

4.4.1 Operational Production Information Systems

There are numerous operational production information systems. Many are part of the financial accounting system of organization. For example, purchasing, accounts payable, inventory, order entry, accounts receivable, and payroll subsystems of the accounting system provide information to support manufacturing and production activities. The next section briefly describes some major operational production information systems.

Purchasing Information Systems

To produce goods and services, you must have the right quantity of raw materials and production supplies on hand. Furthermore, you will want to procure these materials and supplies at the lowest cost and have them delivered at the right time. To assist in this function, the purchasing information system has to maintain data on all phases of the acquisition of raw materials and purchased parts used in production.

Receiving Information Systems

When shipment of purchased goods and supplies are received, they must usually be inspected and verified and the information about their status passed on to the accounts payable, inventory, and production departments. Delivery dates should also be noted so that data on delivery times can be collected. This type of information is supplied by *receiving information systems*.

Quality Control Information Systems

Quality control information systems provide information about the status or production of goods as they move from the raw material state, through goods in process, to the finished goods inventory.

Shipping Information System

At the other end of the production process, there is a need to place finished goods produced into inventory and to ship those on order to customers. There are many records and documents that are used to assist and monitor in the inventorying and shipping processes – for example, shipping reports and packaging slips. The information from the shipping system affects the inventory and accounts receivable systems.

Cost Accounting Information Systems

A variety of operational information subsystems of the financial accounting system collect and report information about the resources that are used in the production processes so that accurate costs of production can be obtained on products and services. Cost accounting systems monitor the three major resources used in production: personnel, materials, and equipment and facilities. Payroll information systems and materials information systems provide information about the costs of these resources.

4.4.2 Tactical Manufacturing and Production Information Systems

Tactical information systems include inventory management and control systems, capacity planning, production scheduling, and product design and development.

Inventory Management and Control Information Systems

The management and control of raw materials, goods in process, and finished goods inventories is an important part of production system. Careful management and control of these inventories will usually provide considerable savings to the organization. Inventory management and control systems use information from operational information systems, such as the shipping and receiving systems, purchasing systems, and order entry systems.

Maintaining inventories at their proper level eliminates production shutdowns from lack of materials and lost sales from lack of finished goods. Ordering too much and ordering too little

are costly in terms of inventory carrying costs and ordering costs respectively. Thus, the best or economical order quantity (EOQ) strikes a balance between carrying costs and procurement costs.

The computation of EOQ for each item in inventory would be a very large and tedious task if done manually. Also, for many inventory items, the manager may wish to play "What if?" formulas games with the values in the EOQ. Without computers, these tasks would be laborious and may prove too time consuming to keep the order process fine-tuned to current data.

Capacity Planning Information Systems

In addition to ensuring that there will be enough raw materials on hand for forecasted production, the production manager must also see to it that there will be enough production capacity available to meet production goals. The purpose of capacity planning is to make certain that there is sufficient personnel, space, machines, and other production facilities available at the right time to meet the organization's planned production. Managers also utilize capacity to minimize capacity in excess of planned production needs.

Production Scheduling Information Systems

The purpose of the production schedule is to allocate the use of specific production facilities for the production of finished goods to meet current or forecasted orders. To manage the scheduling process, a number of scheduling tools have been developed. Two of these are Gantt and PERT charts and there are project software packages at this time.

Product Design and Development Information Systems

Numerous tactical decisions must be made to design and develop a product, especially a new product. The design engineering team usually depends on product specification information derived from customer surveys, target population analysis, or other marketing research information systems. The team also depends on data from the quality control information systems. Teams may use other computerized systems for designing new products as well. Software used for design, testing and manufacture of product is commonly available these days.

4.4.3 Strategic Planning Manufacturing Information Systems

As we just discussed, production information systems are primarily operational and tactical in nature. They are concerned with providing information to monitor and control the production of goods and services and to allocate resources to complete production processes. Manufacturing information systems are typically strategic in nature.

For example, the decision to construct a plant addition or a totally new plant, the selection of the plant site, and the creation of general plans for the design and layout of the facility are top management decisions. Decisions of this magnitude will require the commitment of a large amount of capital and other resources over a long period of time and thus are strategic planning decisions. Such decisions are not made lightly.

The decision to locate a plant site requires a great deal of information that is external to the organization. Facilities planning information systems support top management decisions in this area. Some of the information needed is relatively quantitative ñ for example, the availability and cost of trained or experienced labor and the degree to which it is unionized, the availability and cost of transportation for raw materials and finished goods, the availability of suitable sites and the cost of land, the proximity of raw materials suppliers and/or finished goods customers, the availability and costs of power, and the rate of property and incoming taxation.

Other information used in locating a plant may be qualitative in nature ñ for example, community attitudes toward an organization of the type of wishing to locate there and the quality of community services, such as education and training opportunities.

The final decision concerning plant may be made using some form of weighted average technique, in which factors such as those listed above are ranked and the total scored for several potential sites computed. It may also be made on largely emotional factors by top management.

4.5 HUMAN RESOURCE INFORMATION SYSTEMS

The human resource management function is concerned with the individuals who constitute the organization. From the standpoint of the organization, the function is responsible for the acquisition and effective use of the individual, the function is concerned with the well being, growth, and development of each worker. To achieve these ends, human resource management departments perform a variety of activities. The major activities of the personnel function in an organization include:

1. recruiting employees
2. evaluating applicants and employees
3. selecting, placing, promoting, terminating, and transferring employees
4. analyzing and designing jobs
5. training and development reports
6. producing required governmental reports
7. managing employee wage and benefit plans
8. planning short and long term staffing needs

To perform these activities, managers rely on a number of operational, tactical, and strategic information systems.

4.5.1 Operational Information Systems

Operation human resource information systems provide the manager with data to support the routine, repetitive personnel decisions that occur regularly. There are several operational level information systems that collect and report personnel data. These include information systems pertaining to the organization's positions and employees, and about governmental regulations.

Payroll Information Systems

The financial information system, through its payroll subsystems, collects and reports data pertaining to human resources that is largely operational in nature. Payroll files often contain a great deal of information about employees ñ including information about employees pay rates, wage classifications, and seniority ñ that can be used to managers making human resource decisions.

Position Information Systems

A job is usually defined as a group of like or similar positions. A position, on the other hand, consists of tasks performed by one worker. The purpose of a position information system is to

identify each position in the organization, the job category in which the position is classified, and the employee currently assigned to the position. Reference to the position information system allows a personnel manager to identify the details about unfilled positions. Position information systems also allow the personnel manager to identify human resource problems.

Employee Information Systems

The personnel department must maintain information on each of the organization's employee for a variety of reporting purposes. One part of this information system is a personnel file, which usually contains personal and organization related information such as name, address, sex, marital status, citizenship, years of service or seniority data, education and training, previous experience, employment history within the organization, salary rate, salary or wage grade, and retirement plans.

Another part of an *employee information system* is an employee skills inventory. The skills inventory contains information about every employee's work experience, work preference, test scores, interests, and special skills or proficiencies. The skills inventory system provides information for many decisions.

Employee Evaluation Information Systems

Many organizations review the work of employees on a regular basis to make decisions regarding merit pay, pay increases, transfer, or promotion. Typically, a new employee is evaluated at the end of the first four months and other employees are evaluated semiannually. These reviews provided by the *employee evaluation information systems* are often called *performance appraisals*. The data for performance appraisals are frequently collected by administering employee appraisals forms to each employee's immediate superior. The forms may also be given to peers, the employees themselves, and even customers or clients.

Applicant Selection and Placement Systems

After job and the employee requirements for those jobs have identified and after a suitable pool of job candidates has been recruited, the candidates must be screened, evaluated, selected and placed in the positions that are open. To ensure compliance with federal, state, and local employment laws, these procedures must be fully documented and carried out in a structured

manner. Thus, data pertaining to interviews, examinations, and placement decisions should be collected and kept according to the requirements of the various laws and regulations of acts.

4.5.2 Tactical Human Resource Information Systems

Tactical information systems provide managers with support for decisions that emphasize the allocation of resources. Within the human resource management area, these decisions include recruiting decisions, job analysis and design decisions, training and development decisions, employee compensation plan decision, and labor negotiation decisions.

Job Analysis and Design Information Systems

The information inputs to the job analysis and design information system include data obtained from interviews of supervisors and workers and affirmative action guidelines. Inputs also include information from sources external to the firm, such as labor unions, competitors, and governmental agencies. The outputs of the job analysis information system are job descriptions and job specifications.

Recruiting Information Systems

To develop recruitment plan and monitor its success, a recruiting information system is necessary to collect and process the many different types of information needed to construct the plan. These include a list of unfilled positions; the duties and requirements of these positions; lists of planned employee retirements, transfers or terminations; information about the skills and preferences of current employees; and summaries of employee appraisals. The source of recruitment, affirmative action plans, information about the success of recent recruitment activities must also be included in the information systems.

Compensation and Benefits Information Systems

Fringe benefit is anything in addition to the basic salary and over time pay to an employee. Information on the benefits and compensation packages will help managers to make informed decisions.

Employee Training and Development Systems

To support the management and training function, it is important to keep information about current and potential instructors; current and potential students; the content and scheduling of

courses, seminars, and workshops; and the success of past efforts, as measured by tests, student ratings, and supervisor follow ups. This information system should permit tactical decisions regarding the nature of training and development programs. This system must use the other sources of human resource information systems.

Labor Negotiations Information Systems

Negotiating with the union requires information gathered from many of the human resource information systems already discussed. In addition, negotiators need information from the financial accounting system and from external sources including competitor wage agreements and appropriate economic data for the industry, employee group, and geographical region.

The personnel team completing the negotiating needs to be able to obtain numerous ad hoc reports that analyze the organization's and union's positions within the framework of the industry and current economic situation. It is also important that the negotiating team be able to receive ad hoc reports on a very timely basis, since additional questions and tactics will occur to the team while the negotiations are underway.

4.5.3 Strategic Human Resource Information Systems

Human resource planning is conducted to ensure that the organization has the right kinds and the right numbers of people at the right places at the right time to achieve its objectives. There are two types of human resource planning that are strategic in nature. These are manpower planning, or staffing planning, and program planning.

Manpower Planning

Organizations that are involved in long term strategic planning, such as those planning to expand into new market areas, or to construct new factories or offices in new locations, or to add new products, will need information about the quantity and quality of the work force that is available to achieve their goal. Manpower planning serves this purpose. This type of planning involves identifying the human resources needed to meet the organizational objectives specified in the strategic plan. This means forecasting the supply and demand of the required work force. These forecasts are estimates of the characteristics, quantity, and pricing of the labor force needed to achieve the long-term plans of the organization.

To forecast human resource needs, information will be required to answer a number of planning questions, including the following:

1. What should the labor force of the organization look like to meet the strategic plan? What kinds of skills, experiences, knowledge, and other qualities should the organization's human resources possess? In other words, what are the job descriptions and specifications required by the strategic plan?
2. What quantities of human resource with the qualities already identified are needed to carry out the strategic plan? In other words, how many positions for each job title are needed for the strategic plan?
3. What are the current human resources of the organization and how do they help the organization's strategic needs for human resources?
4. What other human resources are available to achieve the strategic plan?

Identifying the types and quantities of workers needed for the strategic plan is forecasting the demand for human resources. Identifying the human resources available internally and externally is forecasting the supply of those human resources. Forecasting demand and supply can be done on a macroeconomic level or a microeconomic (more specific to the organization) level.

Program Planning

A second strategic planning function is to develop the policies, procedures, and activities that will achieve the human resource needs spelled out in the manpower plan. Two key activities in program planning are job analysis and design and recruiting. Job analysis and design is essential to describe the work force required by the organization.

Recruiting plans must be developed to acquire the right kinds of workers in the right amounts to fill the jobs described. Where there are insufficient workers with the right skills, training programs may have to develop, or the jobs may have to be redefined to fit the skills of the available work force.

Check Your Progress Exercise

1. Define the following terms:
 - A. Financial information systems

B. Marketing information systems

C. Human resources information systems

2. List down at least five (5) sub system modules of accounting systems.

3. Explain the term telemarketing.

4. What is the purpose of manufacturing and production information systems?

5. List some of the major activities of personnel functions in an organizations.

4.6 SUMMARY

In this unit you have learned about the different levels of managers and how information systems support these functions. The activities of an organization include day to day operations, effective allocation of resources, and strategic planning. Information systems are designed to support decisions at these three levels. Operational information systems process transactions describing the acquisition and disbursement of corporate resources. First line supervisors use operational systems to monitor day-to-day activities. Accounts receivable, accounts payable, and payroll systems are examples of operational systems.

Tactical information systems provide middle managers with the information they require to allocate resources needed to achieve organizational goals. Often, these tactical systems summarize or aggregate data generated from operational systems. Strategic planning systems help senior managers identify organizational goals. Information for strategic planning often comes from external sources such as demographic data and competitive industry statistics.

Information systems have evolved within organizations over the past 20 to 25 years. This growth involves the construction of operational, tactical, and strategic planning information systems to support the managers of the business. However, the growth of data processing systems involves more than technology and applications. It requires user involvement in application development, the organization and management of an MIS function, and the development of effective planning and control strategies.

4.7 ANSWER TO CHECK YOUR PROGRESS EXERCISE

1. Refer to sections 7.2, 7.3, and 7.5
2. The sub systems module of accounting includes:
 - a. General ledger

- b. Fixed assets
 - c. Accounts receivable (credit limit, aged accounts receivable reports can be generated)
 - d. Order entry
 - e. Accounts payable
 - f. Inventory
 - g. Purchase order
 - h. Payroll
3. Using the telephone for selling is referred to as telemarketing. Using telemarketing has become a very important means by which companies have improved the productivity of their sales force.
4. ***Purpose of manufacturing & production systems*** encompasses all the activities necessary to ensure production. These activities may include the evaluation of sites for production; the planning, development, and maintenance of production facilities; and the setting of production goals to meet the requirements of the sales forecast generated by the marketing system.
5. The major activities of the personnel function in an organization include:
- recruiting employees
 - evaluating applicants and employees
 - selecting, placing, promoting, terminating, and transferring employees
 - analyzing and designing jobs
 - training and development reports
 - producing required governmental reports
 - managing employee wage and benefit plans
 - planning short and long term staffing needs

Unit Five

Information Resource Management

5.0 AIMS AND OBJECTIVES

After studying this unit, you should be able to:

- better understand the structure of the information systems
- explain the major functions of information systems
- know how managing system development effort and its technology looks like
- describe the management of system operations and
- Understand managing end users computing.

5.1 INTRODUCTION

Computer information systems are often large and complex. With the advent of minicomputers and personal computers, computer systems are being dispersed to a large number of points at far-flung locations. Computer technologies such as electronic mail, decision support systems, and personal computers, application development by users, computer networks, and computer integrated manufacturing are causing monumental changes in the conduct of business activities. Managers who successfully manage these technologies will be winners in the competitive business arena.

Management of the information systems resource has been significant changes in recent years. In the 1960s and 1970s, information systems managers gradually increased their power and influence within most organizations. Typically, they moved from managerial positions within another function, such as accounting, to vice-presidential positions within the information systems function. These managers were looked upon as experts in information processing, and they held the keys to the computer resource. In recent years, distributed data processing and application development by users have resulted in a challenge to the power of the information systems manager. With users directly purchasing hardware and either purchasing or creating software, many see the role of the information systems manager changing to that of a consultant, an advisor, and a coordinator. However, centralized computer facilities will continue to exist, especially for large databases that support many users concurrently. The challenge of administering centralized database and communication networks with distributed computing will continue to provide information systems managers with a crucial role in the management of business organizations.

This unit first discusses the structure of the information system function. It then looks at managing system development. Finally, it covers the management of system operations,

including system maintenance, data processing (DP) operations, the physical security of computer operations, and managing end-user-computing.

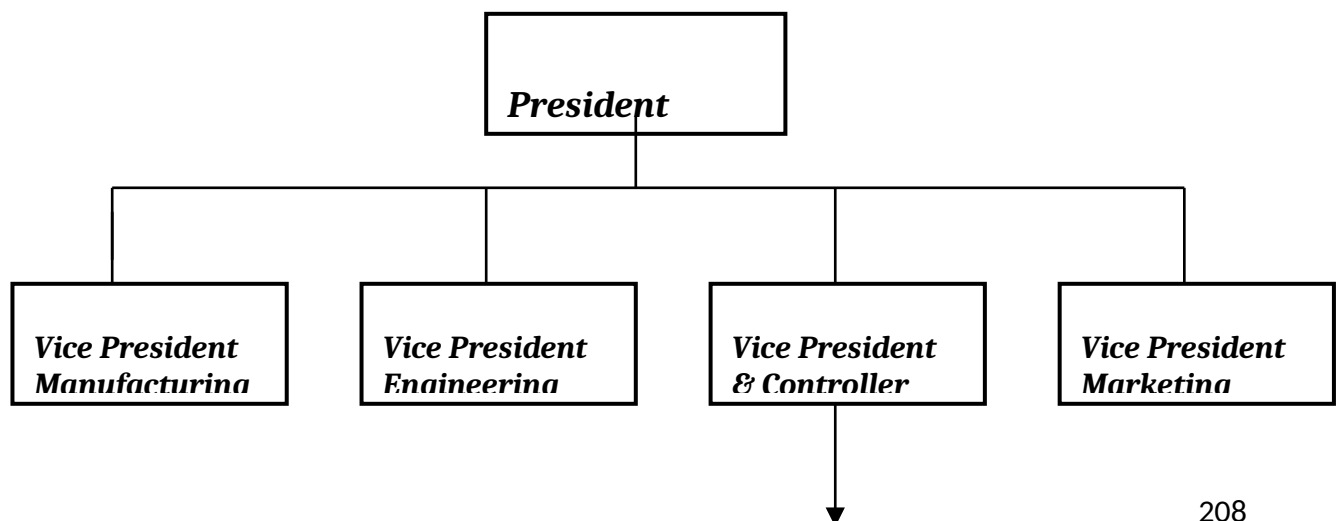
5.2 STRUCTURE OF THE INFORMATION SYSTEMS FUNCTION

This examination of the management information systems function is broken down into two major areas: the location of the information systems function within the structure of the organization, and the internal structure of the information systems function itself.

5.2.1 The Organizational Location of Information System

Typically the information systems function is located in one of two areas in an organization. The chief information systems executive may report to the vice president and controller as depicted in Figure 6-1, or he or she may report directly to the president as vice president and chief information officer, as depicted in Figure 6-2. There are advantages and disadvantages to both of these organizational locations for the information systems functions.

The **controller**, being the chief financial officer in a corporation, is looked upon as the primary provider of financial management information. In addition, functions in the controller's areas such as payroll, accounts payable, and accounts receivable are often the first applications to be computerized.



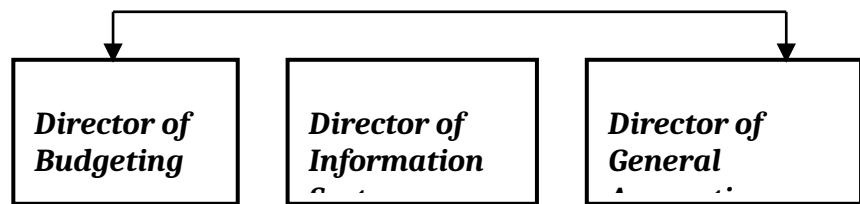


Figure 5-1: Chief information systems

Therefore, the information systems function often originates and matures within the controller's organization. The primary disadvantage of having the chief information systems executive report to the controller is that the computer resource may be dominated by and used primarily to solve problems within the accounting area. Therefore, other functions such as marketing, engineering, and production applications of the computer may be neglected. The primary advantages of having the chief information systems executive report to the controller are that accounting is an information-oriented discipline and accountants are well trained in the area of control. These skills of accounts may produce a computer system that is well controlled and easy to audit.

Many large, mature information systems organizations are separate and have a chief information officer who is a vice president and reports directly to the president, as illustrated in Figure 5-2. This location for the information systems function helps ensure that each functional area receives unbiased attention from the information systems department. Today, with computers penetrating many aspects of all functions, it is particularly important that the chief information systems executive be a member of the unbiased vice presidential level of management.

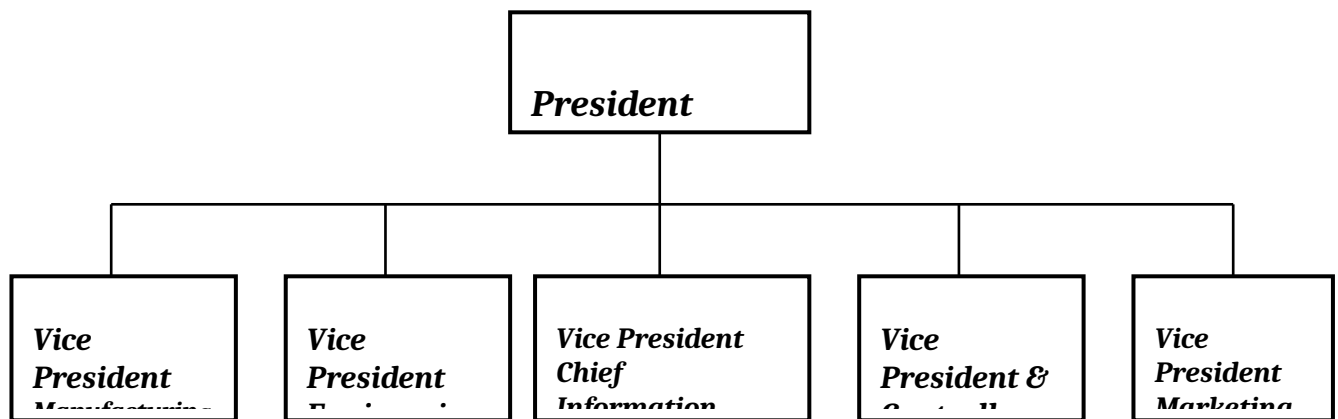


Figure 5-2 Chief information systems executive reporting to the president

5.2.2 The Internal Structure of Information Systems

The organization of the information systems function itself varies from firm to firm, particularly between small and large firms. Figure 6-3 depicts a typical information system organization in a large firm. In small organizations, many of the functions identified in this figure are consolidated or do not exist. For example, a small firm may purchase all of its software, so it has no need for the systems and programming function and the systems software function.

As shown at the manager level of Figure 5-3, six distinct functions should be carried out within information systems or arranged through outside sources. A specific information systems organization chart may look quite different from Figure 5-3. However, the major functions that an information systems organization is concerned with are covered in the figure.

The **information center** is responsible for coordinating and supporting application development by end users. The people in the center are technical experts on the software that users typically employ without programming, such as Lotus 1-2-3 and FOCUS. They act as consultants to assist the user in application development without programming. Since the personal computer is used in this effort, the coordination of personal computing is often carried out by the information center. The personal computer has brought several challenges, including managing the personal computer. The responsibilities of the personal computer coordination group are listed in Figure 5-4.

The **systems software department** installs and maintains system software such as operating systems and database management systems. The people in this department are technical and very highly skilled programmers who rarely deal directly with users.

Application software is developed by, or selected and purchased with the help of, the **systems and programming department**. The people in this department interact heavily with users as they develop users applications.

The **technical support staff** is in charge of maintaining hardware and establishing data processing standards. These standards are very much like the procedures for the data

processing function. They include such things as program, data, and application naming conventions, procedures for maintaining the integrity of communication systems, and guidelines that govern the content of user procedure manuals.

The data processing operations department manages the day-to-day operations of the computer hardware. Employees of this department also monitor the processing of computer jobs and assist when human intervention, such as mounting tapes, is required.

A function that is relatively new in most large information systems organizations is that of database administration (DBA). The DBA department is responsible for coordinating the database and providing for data security.

The information systems steering committee is made up of high-level managers from each function within the business, including marketing, accounting, information systems, manufacturing, and so on. Its purpose is to guide the overall direction of information systems. For example, the steering committee decides the priorities for implementing specific application systems. Much of the high level planning for information systems is either performed or approved by the steering committee. Ideally, the steering committee provides the broad perspective and guidance necessary to ensure that the information systems resource supports the objective of the business as a whole.

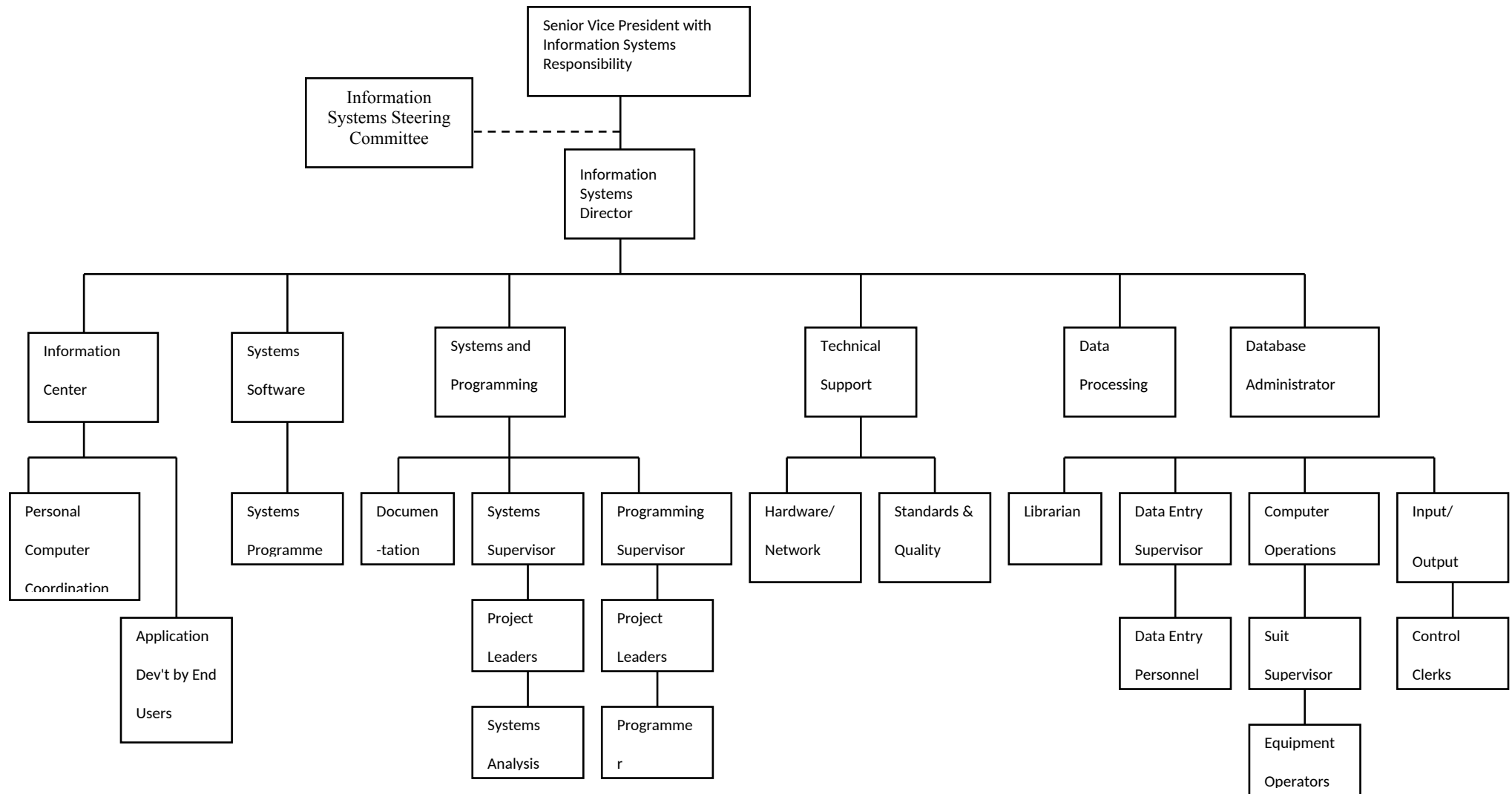


Figure 5-3: Large information systems organizations

Figure 5-4 Responsibilities of the personal computer coordination group

- Standardizing hardware purchases
 - Standardizing software purchases
 - Providing PC access to mainframe databases
 - Preventing the redundant development of software
 - Preventing the redundant creation and storage of data
 - Assuring that data stored on personal computers are secure
 - Providing personal computer training
 - Maintaining the firm's in-house "personal computer store" where users can try out a wide variety of PC hardware and software.
-

5.3 MANAGING SYSTEM DEVELOPMENT

Perhaps the most important aspect of managing the system development effort is the **system development methodology** used. A structured methodology, blended with system prototyping and application development by users, should be used to develop any new system.

Another aspect of managing system development is the process by which development efforts are controlled. Most system development organizations use a project management approach. Under the project management approach, each application development of significant size is assigned to a project development team. This team is usually headed by a senior systems analyst, or sometimes by a user department manager, who has system development experience. Sufficient resources in the form of programmers, systems analysts, hardware, and software are assigned to the team to complete the project. The team approach to management is being applied to all areas of information systems management.

Each project is assigned its own financial budget and time schedule. The financial budget performance is tracked by periodic reports, which compare actual expenditures to budgets. Schedule performance is managed and controlled through a project schedule performance tool such as **program evaluation and review technique (PERT), critical path management (CPM), or Gantt charts**. PERT and CPM are scheduling methods that use networks consisting of activities that consume resources and take time, and events that mark the beginning and end of

each activity. These methods allow the minimum amount of time in which a project can be completed and the critical path to be determined.

A Gantt chart can be a valuable measurement tool in complex project. It aids in scheduling and coordinating and provides a visual means of evaluating progress. Since preparing a Gantt chart does not require extensive effort or data, the potential benefits generally exceed the cost.

5.4 MANAGING SYSTEM OPERATIONS

In addition to managing system development, the director of information systems is responsible for managing the operation of information systems. This task involves overseeing system maintenance, data processing operations, and the physical security of the system.

System Maintenance

System maintenance is the correction of errors discovered in programs and the modification of programs to satisfy changed user requirements or conditions. Changes in programs are often also necessary when new hardware is introduced. System maintenance has become a challenge for many information systems organizations. In many cases it consumes 70 to 80 percent of the systems and programming resources. Such situations leave few resources for developing new systems. Methods for decreasing the cost of system maintenance are of paramount concern to information systems managers.

As with managing systems development, the adoption of a structural system development methodology greatly enhances the ability to manage system maintenance. Efficient and effective system maintenance requires an understanding of the program to be modified. Programming personnel must be able to effect the change by changing the program statements in a confined and isolated area of the program. If a change requires modification in many different areas of the program or system, then system maintenance becomes an almost impossible task. Structured methodologies produce modular programs. Each module is an independent and self contained as practical. Therefore, changes are likely to affect only a restricted area of one module.

Often organizations have programs that meet their needs but are difficult to maintain because they are non-modular or they use unstructured code. These organizations may hire firms that specialize in rewriting programs to restructure these programs in a modular form that follows structured programming standards. Some programs are rewritten manually, but COBOL

programs are often restructured using other programs that do not require human intervention. Rewriting can often extend the life of a program for several years.

Active data dictionaries also simplify system maintenance by restricting both the number and location of program changes. **Active data dictionaries** allow data to be defined in one location. The data definitions are then used by all programs that process those data. The programs use each data item by referring to a data name. The physical format of the data, including their type (character or numeric) and their length, must be known to each program that uses them. Thus, the physical format definition of a data item is also stored in the active data dictionary. The term active is used since each program actively uses the data dictionary for data definitions.

Requests for changes in programs originate with users and should be made on a formalized change authorization form. On this form the user identifies the program or system to be changed and outlines the changes desired. This form requires authorization signatures, which must be obtained before the changes are made. Authorization signatures are typically required from user management and from the systems and programming manager, and are sometimes required from the data processing steering committee. Figure 8-5 shows a change request and authorization form.

Actual changes to programs must be well managed and controlled. After the changes have been made, the system is thoroughly tested using the same set of tests used when the system was developed. The tests are modified to reflect changes and are then run before the changed system is implemented.

Data Processing Operations

Managing data processing operations is much like managing any production shop within an organization. Management is concerned with maintaining sufficient capacity to process the computer jobs. Users of the resource are charged for the resources they use. Personnel are hired, managed, and sometimes dismissed, and the machines are maintained in operable condition.

REQUEST FOR SYSTEM MODIFICATION
User Name:
Department:
Telephone No.:

System Name:			
Module(s) to be changed (if known):			
Please describe the change(s) desired and explain the reasons thereof:			
Signature	Approved	Rejected	Comments
Requestor	☐	☐	
Dept. Manager	☐	☐	
IS Liaison Officer	☐	☐	
Manager, Systems and Programming	☐	☐	

Figure 5-5 Change requests in an authorization form

Processing capacity may be limited by any number of factors including primary storage size, secondary storage size, CPU power, number of terminals, and so on. Any of these can become a **bottleneck** that limits the capacity of the computer system. Data processing management monitors these resources and determines if one is likely to become a bottleneck in the future. Additional resources usually can be obtained at a reasonable price if the potential capacity bottleneck is identified promptly. Software monitors such as IBM's system management facilities help determine the usage levels of various system resources. For example, managers can determine the percentage of time various terminals are used and the time of the day they are used. They can also determine if the CPU is running close to its maximum capacity at any given point during the day. Trends of system resource utilization enable managers to project when various resources will be used at their capacity and, therefore, when expansion should be planned. Hardware monitors are also sometimes used to detect bottlenecks and to determine utilization levels for various devices.

An effective way to ensure that the computer resources is used efficiently is to charge users for their use of the various computer resources including CPU time, disk space, tapes, printing, and son on. Under a **user building system**, the DP operations department is set up as a service center to the rest of the firm. Its services are available to anyone in the firm who is willing to pay for those services. Rates for the services should be similar to the rates charged if the user contracted for data processing services outside the firm with, for example, a service bureau.

Physical Security

In many companies today the information systems resource is a crucial asset. Even a temporary loss of this resource through fire, sabotage, or other disaster can be costly. You will likely soon learn firsthand the need for contingency planning, with backup of files. After you have worked for several days on an electronic spreadsheet, program, or word processing file, you will accidentally lose or erase it. Each term, one or more of my students indicate that they have lost their boxes of disks. Typically, many days of work are stored on those disks, and they have not been backed up. One such experience usually teaches the importance of planning for disasters. Imagine the cost to a large business if it lost all its files in a fire and no file backups had been made!

File backup is a technique for recovering from a disaster after it occurs. This section examines physical security that helps prevent disasters from occurring. It covers five areas within physical security: entry, sabotage, fire, natural and environmental disaster, and power controls.

Entry Control: A well designed entry control system controls entry to the computer facility. Only operations personnel are allowed to enter the computer facility itself. Programs and data that are manually delivered to the computer room are passed through a window. (Most programs and data today are transmitted electronically to and from the computer, and thus require input data and program change control rather than entry control). **Entry control** is usually done through locked doors to the computer room. These doors are opened through various means, such as a plastic card with a magnetic strip similar to a credit card, or a combination of both a plastic card and a memorized entry code.

Plastic cards and key can be lost or misplaced, and thus may be used by someone other than the person authorized to use them. Providing 100 percent positive identification of each person entering a computer facility is difficult. Firms have tested, and are using, a wide variety of

techniques and machines to provide positive identification. One machine examines the pattern of each person's fingerprints. (You cannot misplace your fingerprints!) But it has been found that a photocopy of a person's fingerprints can trick the machine into unlocking the door. Another machine works similar to the fingerprint machine, except each person has to kiss the machine! Lip prints also uniquely identify a person. But aside from its obvious drawbacks, this machine can also be misled by a photocopy of a person's lip prints. A third machine shines light into each person's eyes and identifies him or her based on the retinal patterns within the eyeballs. Although this technique is harmless, employees probably would not accept it. Eventually, though, machines that are highly reliable in identifying individuals will be developed.

Sabotage Control: physical sabotage of the hardware, programs, and data is prevented to a large extent through passwords and by physical entry controls over the computer facility itself. In addition, firms that specifically design a secure computer facility usually provide **sabotage control** by constructing the computer room to bombproof specifications. When computers were first used in the 1950s and 1960s, it was common to have large glass windows in the walls of the computer center so that a firm could show off its computer facilities. A saboteur could have easily tossed a bomb through those windows.

Fire Control: the most likely physical threat to a computer facility is fire. The best **fire control** procedure is to store backup copies of data and programs at another location and to arrange for emergency use of alternative computer hardware in case of a fire. In addition, many computer centers use a fire suppression gas known as halon. The halon is released by fire and smoke detection systems, and it is effective in suppressing fires. The primary disadvantage of halon is that it is very expensive. An accidental release of the gas may cost a firm several thousand dollars just to replace the gas. Other, less expensive gases are available, but they are impractical because they are poisonous to operations personnel. Halon is a nonpoisonous gas.

Natural and Environmental Disaster Control: natural and environmental disasters should be considered when choosing the site for a computer room. **Disaster control** provides protection against such catastrophes as floods, hurricanes, bursting pipes, and so on. Water can destroy the sensitive electronic equipment in a computer room. Fire control water sprinklers on floors above the computer room are an environmental hazard that is sometimes overlooked. Water leaking down through pipe holes and other crevices onto computer equipment, tapes, and disks can be

very damaging. For this reason, the floor above the computer room should be thoroughly sealed to prevent water leakage.

Power Control: large computer systems should have uninterruptible and controlled power systems. Plugging such computer systems directly into electrical lines is not a good idea. If the power goes off, even momentarily, the data and programs stored in semiconductor primary storage are lost because semiconductor storage is volatile. Furthermore, power spikes (increases in the current voltage) can heavily damage a computer system. Power spikes are often caused by electrical thunderstorms. Some computer centers guard against these types of power spikes by shutting down computer operations during electrical thunderstorms. However, in many situations shutting down the computer during electrical thunderstorms is not a viable alternative.

Many organizations use power supply systems to provide **power control**. These systems consist of batteries and backup generators. The batteries are continuously charged by the incoming electrical service, and the computer draws its power from the batteries. The computer is insulated by the batteries from the electrical service lines, and can thus avoid power spikes. In the case of a power outage, the batteries are sufficient for a short duration. Long power outages are covered by the backup generator system.

Personal computers can be protected against power spikes and electrical noise with inexpensive devices that shut the power off when these electrical disturbances occur. Uninterruptible power systems can also be purchased for PCs, but they are expensive.

5.5 MANAGING END USER COMPUTING

In this unit we will cover some of the concepts surrounding the primary issue in end-user computing: How do you divide the responsibilities for end user computing between the end user and the central information systems organization? In the early days of personal computers and end-user computing there was very little control in management of end user computing. In fact, some argued that end user computing should be managed with a laissez-faire approach; that is, end users should be free to purchase and use whatever personal computer hardware and software they deemed necessary without any management from the central information systems organization. The arguments for a laissez-faire approach revolved around the contention that end users best knew what hardware and software suited their tasks and that to control end-user computing would stifle the creativity and innovation of end users.

Most experts today would agree that a laissez-faire approach to end user computing is the prescription for a disaster consisting of duplicate systems, incompatible systems, and corrupt and lost data. Thus, in addition to the support and consulting that the central information systems organization should provide to end users, the central information systems organization and end users must arrive at some division of the responsibilities for end user computing. There are potential negative consequences if the central information systems organization dominates end user computing and there are also negative consequences if end users dominate end user computing.

The particular responsibilities, that information systems and users will have depends on the individual organization. However, we can list the minimum responsibilities of each group. Central information systems should have responsibility for the following:

1. Large multi-user databases
2. Information center consulting
3. Implementing and maintaining the network
4. Procedures to compare the costs of internal development of systems projects versus outsourcing
5. A list of preferred hardware and software and their suppliers
6. A guide and detailed checklist for the questions that should be asked when purchasing any hardware or software
7. An inventory of all information systems hardware and software
8. A plan for the career paths of information systems professionals throughout the organization
9. An information systems master plan that outlines the strategic plan for information systems development

As a minimum users should have the following responsibilities to fully participate in all information systems development and maintenance efforts that affect them: To receive, review, and approve all expenditures of funds for information systems that are being developed or are in operation for the user departments' benefits.

As organizations becomes more dependent on computer systems, managing information system resources become more critical. Many of you will be involved in varying degrees in information systems resource management. Sometimes we start to believe that to manage information

systems we must stay on the cutting edge by adapting all the latest technology. It is important to remember that many firms are successful with a more conservative approach.

Check Your Progress Exercises

1. What are the two possible locations for an information system department within the firm?
How do they differ?

2. Identify the major functions within an information systems organization.

3. What are the basic tasks for managing the operations of information systems?

4. List out the five areas that physical security covers.

5. What are the minimum responsibilities of central information systems and of end users in the management of end user computing?

5.6 SUMMARY

- In recent years the role of the information systems manager has changed as users have become more actively involved in data processing.

- The information systems function may be located within the controller's organization where the controller is considered to be the primary provider of information. On the other hand, many organizations treat information systems as a separate function with a vice president who reports directly to the president.
- A structured system development methodology should be used. Normally system development is carried out using the project management approach for project control. If a structured methodology is used for system development, program maintenance becomes an easier task. However, all changes must be authorized and properly documented in order to prevent confusion and chaos.
- Active data dictionaries substantially reduce the number of program changes that must be made when the format of data changes.
- Operations should be constantly monitored to detect bottlenecks and inefficiencies. User billing should be used to ensure efficient use of resources.
- Physical security of the computer system is a major responsibility of systems management. Procedures should be implemented for both the prevention of disasters and recovery from disasters like fire and flooding.
- Determining the responsibilities of central information systems versus end users is one of the key questions in the management of end user computing.

5.7 ANSWER TO CHECK YOUR PROGRESS EXERCISE

1. ° Chief information systems executive reporting to the vice president.
 ° Chief information systems executive reporting to the president.
 For more on this refer to part 6.2
2. ° The organizational location of information system and
 ° The internal structure of information system functions
3. ° System maintenance

- ° Data processing operations
 - ° Physical security of the system
4. Entry, sabotage, fire, natural & environmental disaster and power controls.
5. ° Large multi-user databases
- ° Information center consulting
 - ° Implementing and maintaining the network
 - ° A list of preferred hardware and software and their suppliers
 - ° An inventory of all information systems hardware and software Ö
- For more detail on this refer back to part 8. 5